

Human Health and Disease

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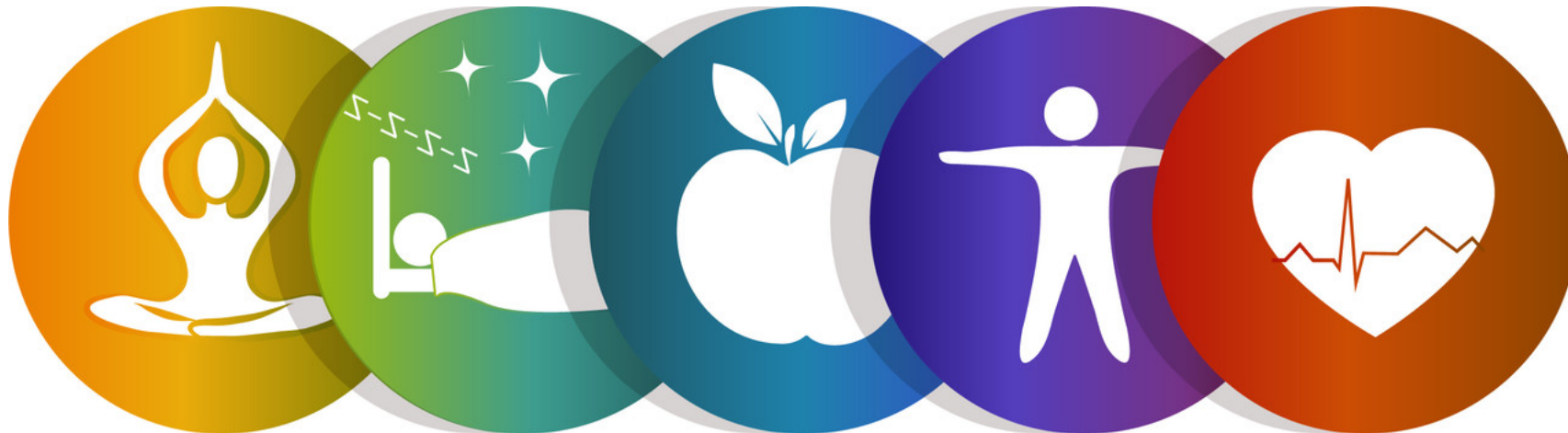
Lecture 1, BT1010



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Health

- State of complete social, mental and physical well being
- Balanced diet, personal hygiene and regular exercise are very important to maintain good health



What helps in achieving good health?

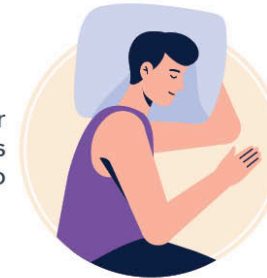
- Awareness about diseases and their effect on different bodily functions.
- Vaccination (immunization) against infectious diseases.
- Proper disposal of wastes.
- Maintenance of hygiene in food and water resources.



Eat a balanced diet of fruits, vegetables, and proteins



Exercise about 30 minutes most days



Aim for 7-9 hours of sleep



Stay connected socially



Reduce stress by prioritizing self-care



Maintain your brain by learning new skills



Schedule routine checkups with your provider

What affects health??

- Genetic disorders – deficiencies with which a child is born and deficiencies/defects which the child inherits from parents from birth
- Infections
- life style including food and water we take, rest and exercise we give to our bodies, habits that we have or lack etc



- When the functioning of one or more organs or systems of the body is adversely affected, characterized by appearance of various signs and symptoms, we say that we are not healthy, i.e., we have a **disease**

Classification of disease

Infectious

Easily transmits from one person to another
Examples: AIDS,

Non-Infectious

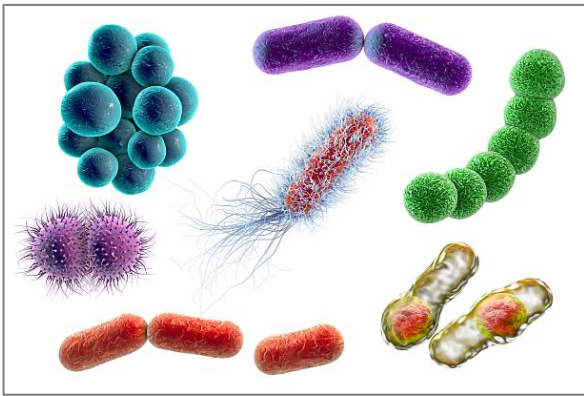
Do not transmits from one person to another
Example: cancer

Pathogens:

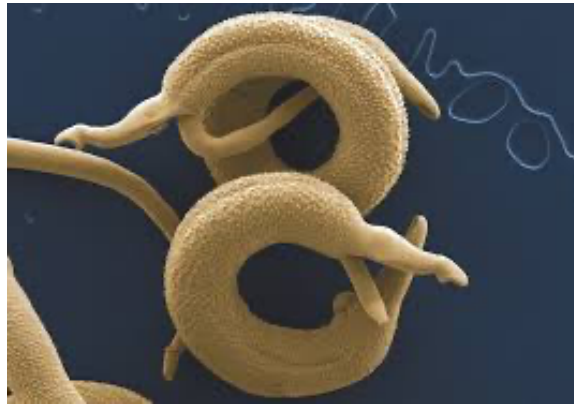
Diseases causing organisms are called pathogens.

Types of Pathogens:

Bacteria



Helminths



Fungi



Protozoans



Virus



Common infectious diseases

Bacterial diseases

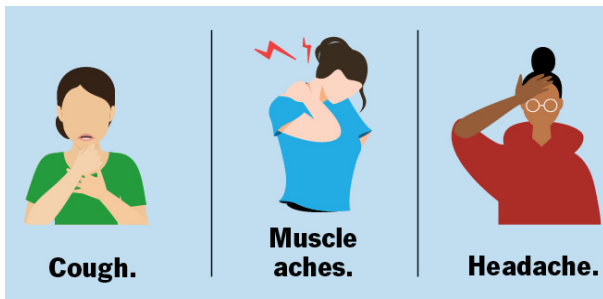
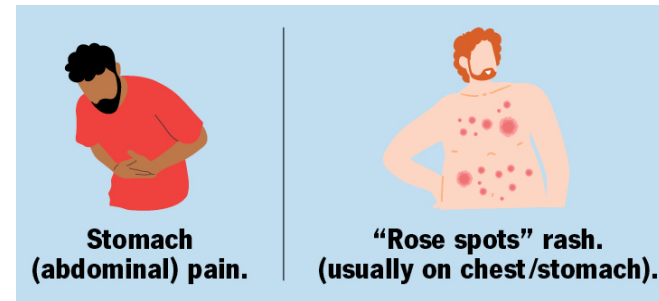
Typhoid fever *Salmonella typhi*



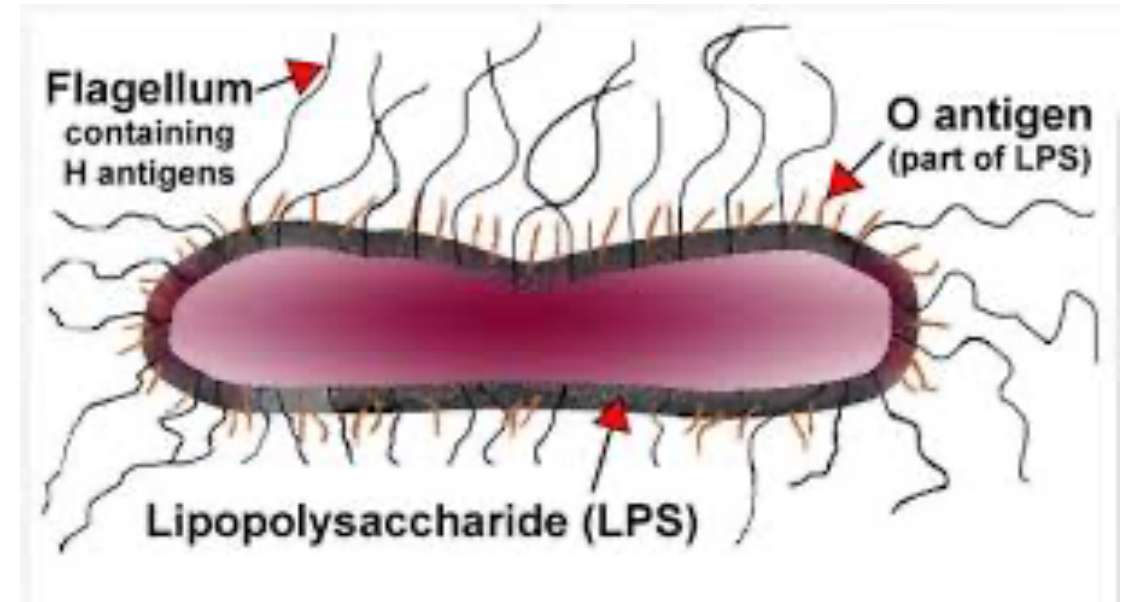
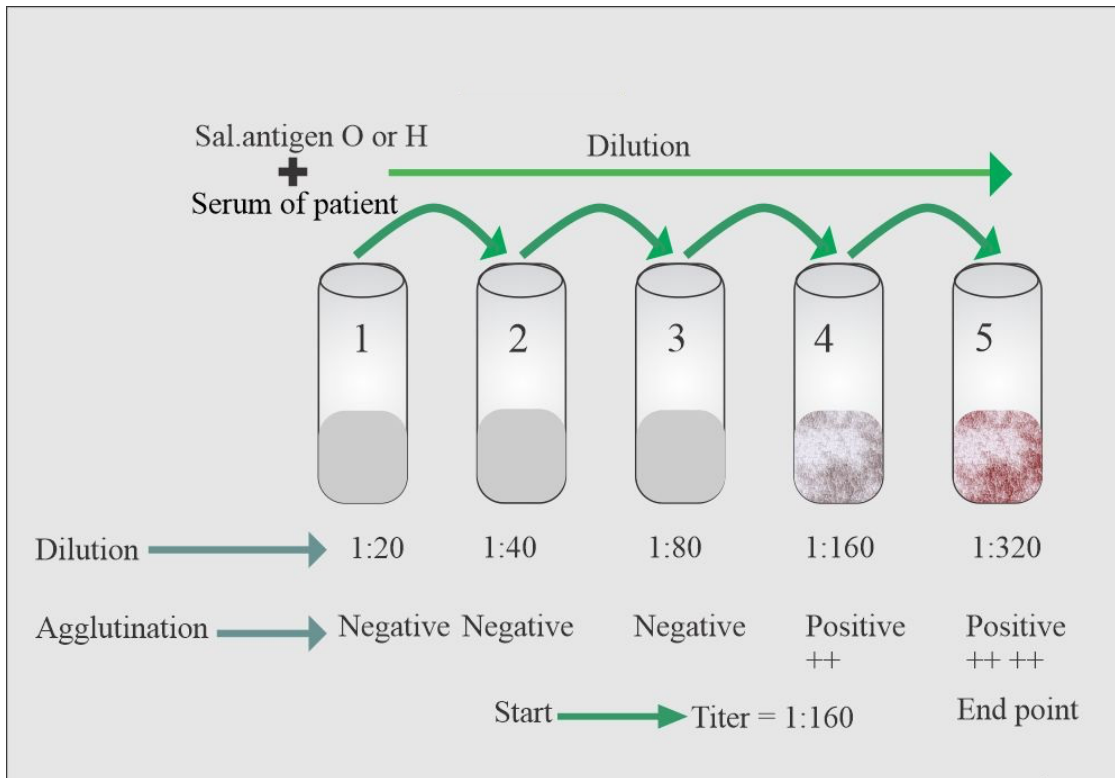
Pathogen: *Salmonella typhi*

Mode of transmission: It enters small intestine through food and water and migrates to other organs through blood

Symptoms: High fever (39-40 °C), headache, weakness, stomach pain, constipation and loss of appetite



WIDAL test is used for the confirmation of Typhoid fever



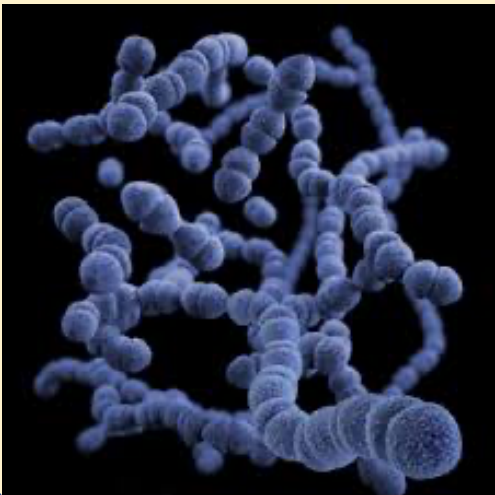
Agglutination: Clumping of particles together. Aggregation of antigens and antibodies.

Mary Mallon nicknamed **Typhoid Mary** was a professional cook. She was a **Typhoid carrier** who continued to spread typhoid for several years through the food she prepared.

Bacterial diseases

Pneumonia

Streptococcus pneumoniae



Pathogen: *Streptococcus pneumoniae*, infect the lung alveoli (air filled sacs of lungs).

Mode of transmission: A healthy person acquires the infection by inhaling the droplets/aerosols released by an infected person or even by sharing glasses and utensils with an infected person.

Symptoms: fever, chills, cough and headache, In severe cases, the lips and finger nails may turn gray to bluish in colour.

Other bacterial diseases:

Dysentery, plague, diphtheria

Viral diseases

Common Cold
Rhino Viruses



Rhinovirus

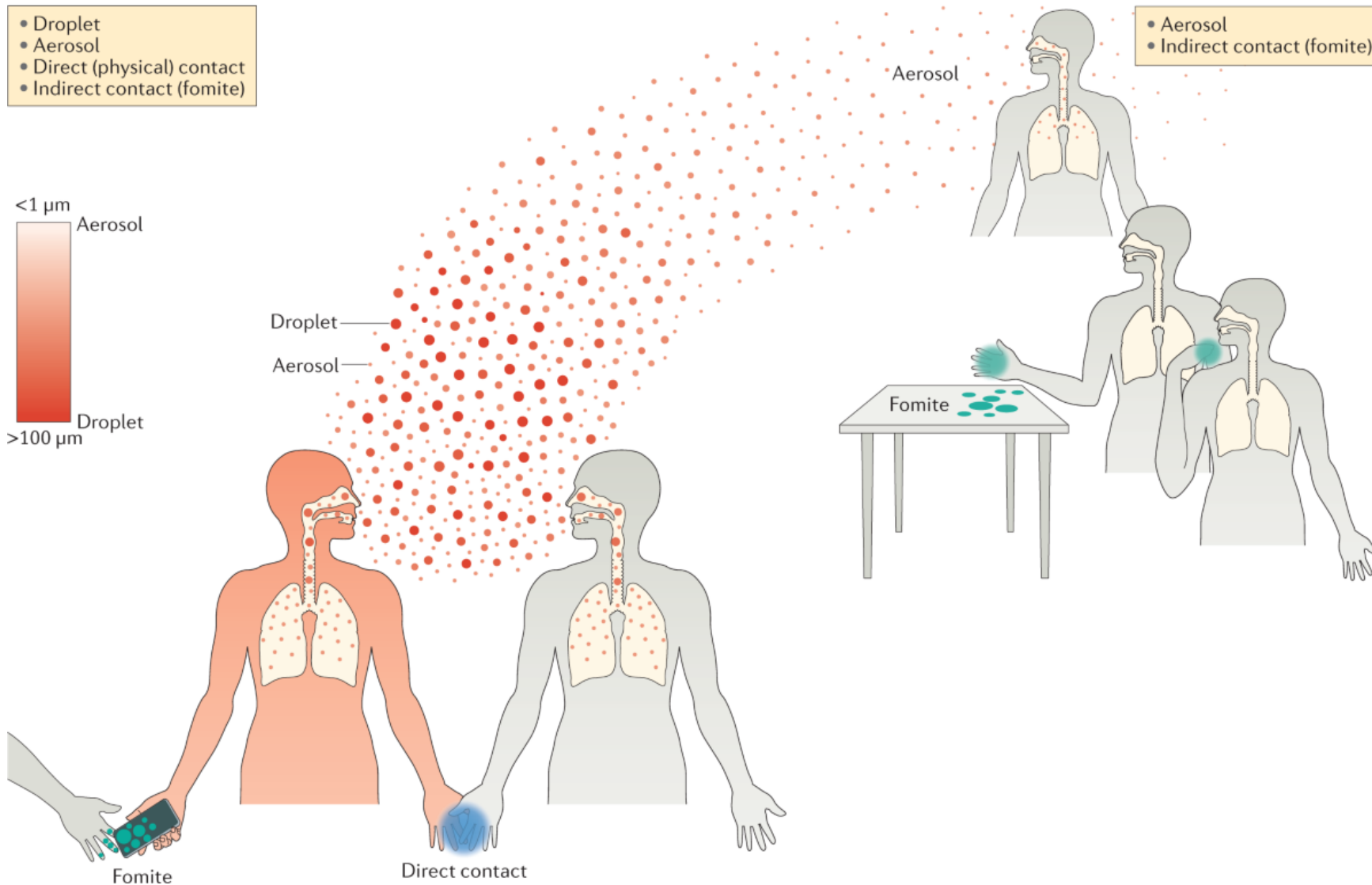
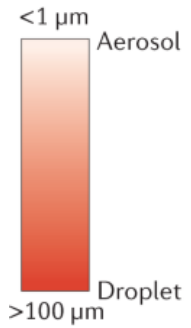
Pathogen: Rhino Viruses, **infect** the nose and respiratory passage but **not lungs**.

Mode of transmission: Droplets resulting from cough or sneezes of an infected person are either inhaled directly or transmitted through contaminated objects such as pens, books, cups, doorknobs, computer keyboard or mouse, etc.,

Symptoms: nasal congestion and discharge, sore throat, hoarseness, cough, headache, tiredness, which usually **last for 3-7 days**.

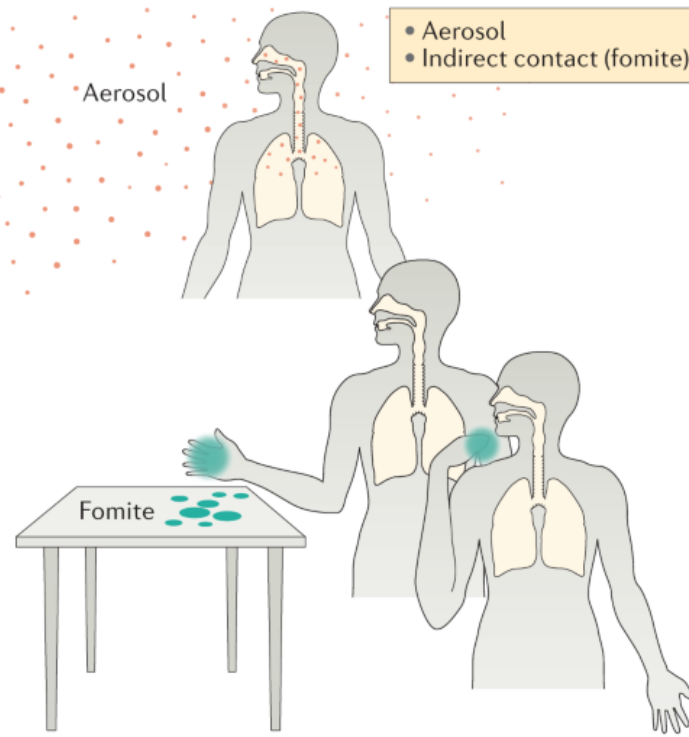
Short-range transmission

- Droplet
- Aerosol
- Direct (physical) contact
- Indirect contact (fomite)



Long-range transmission

- Aerosol
- Indirect contact (fomite)

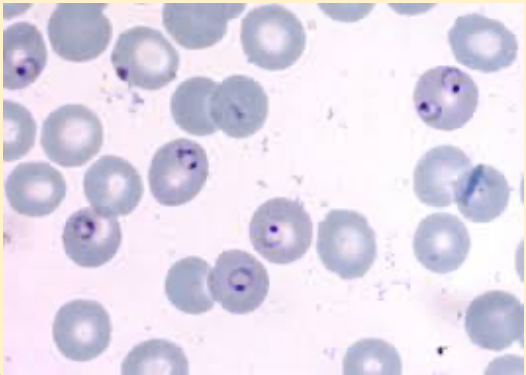


Fomite: Objects or materials which are likely to carry infection, such as clothes, utensils, furniture, pens etc.

Protozoan diseases

Malaria

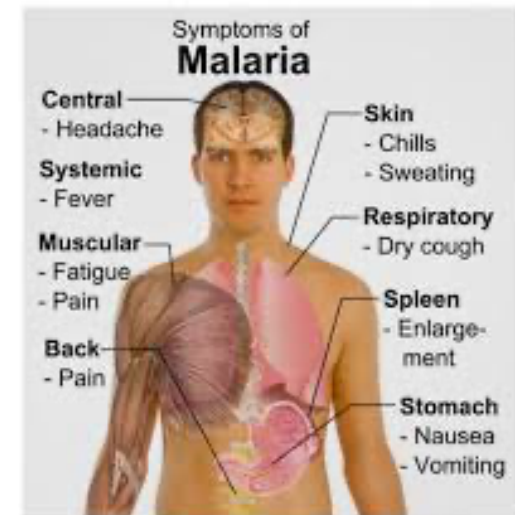
Plasmodium sp.



Pathogen: Plasmodium sp. (*P. vivax*, *P. malariae*, *P. falciparum*). *P. falciparum* causes most serious malaria

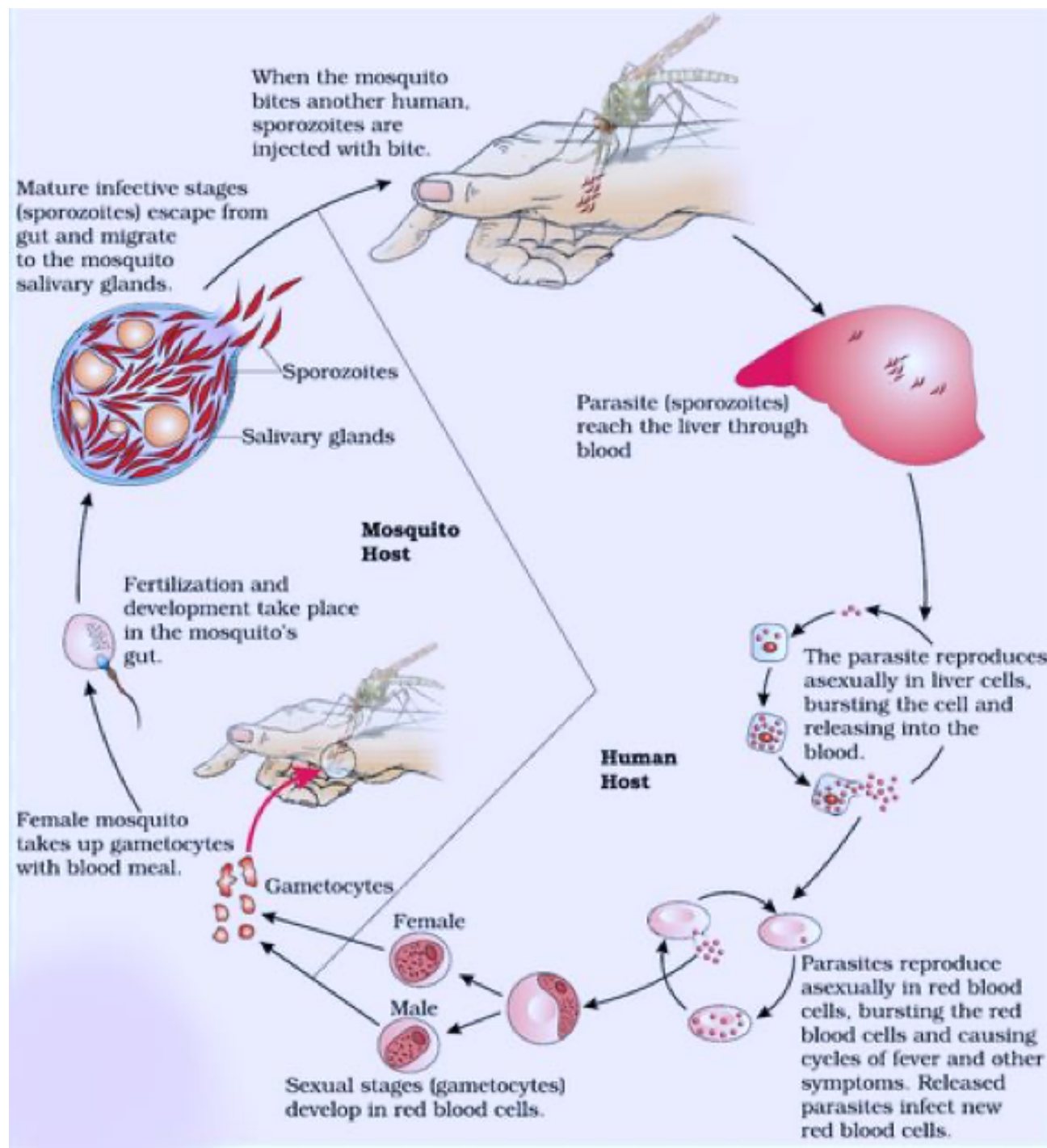
Mode of transmission: by biting of female *Anopheles* mosquito.

Haemozoin (toxin released by Plasmodium) causes chill and high fever recurring every 3-4 months.



Life cycle of Plasmodium

Sporozoites: Mature infective form of Malaria parasite



Protozoan diseases

Amoebiasis (Amoebic dysentery)



Pathogen: *Entamoeba histolytica*

Mode of transmission: Houseflies (mechanical carriers) transmit parasites from faeces to food and water.

Symptoms: Constipation, abdominal pain and cramps, stools with excess mucous and blood clots.

**KEEPING
FLIES AWAY
FROM FOOD**



Helminth diseases

Ascariasis



Pathogen: *Ascaris* (Intestinal parasite)

Mode of transmission: soil, water, vegetables, fruits etc. contaminated with feaces containing eggs of parasites.

Symptoms: Internal bleeding, muscular pain, fever, anemia and blockage of intestinal passage.

Helminth diseases

Pathogen: *Filarial worms* or *Wuchereria* (*W. bancrofti* and *W. malayi*)

Mode of transmission: bite of female **Culex** mosquito

Symptoms: Filarial worms live in lymphatic vessels (usually of lower limbs). It causes chronic inflammation of the organs in which they live for many years. Limbs and genital organs may be deformed.

Filariasis
(Elephantiasis)



Fungal diseases

Ring worms

Pathogen: *Microsporum*, *Trichophyton* and *Epidermophyton*. They are seen in groin, between toes etc.

Mode of transmission: from soil or by using towels, cloths, comb etc. Heat and moisture help fungi to grow.

Symptoms: dry, scaly lesion on skin, nail, scalp etc. intense itching .



Prevention and control of diseases

- Personal Hygiene

- Keep the body clean.
- Use clean drinking water, food etc.



I KEEP CLEAN



Brush your teeth



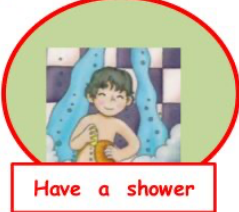
Wash your face



Wash your hands



Comb your hair



Have a shower

- Public Hygiene

- Proper disposal of wastes and excreta
- Periodic cleaning and disinfection of water reservoirs, pools, cesspools and tanks
- Avoid contact with infected persons or their belongings (to control airborne diseases)
- Standard practices of hygiene in public catering.
- Control and eliminate vectors (mosquitoes)

- Vaccines and immunization helped to control diseases like smallpox, polio, diphtheria, pneumonia and tetanus.
- Drugs like antibiotic helped to treat infectious diseases.



A 3D illustration of a human figure, rendered in a light blue, almost translucent style, standing against a bright blue background. The figure is surrounded by numerous colorful, stylized microorganisms. These include various shapes of bacteria (rod-shaped, spherical, and spiral), viruses (spherical with spikes or smooth surfaces), and other cellular structures. The microorganisms are in various colors like red, green, yellow, and blue. The overall composition suggests a theme of human health, microbiology, or the immune system.

Immune system:

It is the system that gives immunity to the body by recognizing, responding and remembering foreign antigens.

Immunity:

Immunity: Derived from word *immunis* 'exempt': A state of protection from infectious disease.

Early vaccination studies led to the immunology: **Edward Jenner** (Small Pox)



- Immunity plays a role in allergic reaction, auto-immune disease and organ transplantation.
- It includes lymphoid organs, tissues, cells and antibodies

